

Ponnu Soman | Data Analyst

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PROFESSIONAL SUMMARY

Data Analyst with around 3 years experience in fraud detection, customer analytics, and data-driven decision-making. Proficient in SQL, Python, Excel, and Power BI for analyzing large-scale datasets, uncovering patterns, and building interactive dashboards. Experienced in feature engineering, predictive modeling, and cloud-based data pipelines using AWS and Azure. Demonstrated impact in reducing fraud losses, improving data quality, and enhancing business insights through scalable analytics solutions across financial services and consulting environments.

TECHNICAL SKILLS

Programming Languages: Python (pandas, NumPy, scikit-learn), R
Data Analysis: Exploratory Data Analysis (EDA), Statistical Analysis, Data Cleaning
Machine Learning: Supervised & Unsupervised Learning, Regression, Classification, Clustering
Data Visualization: Power BI, Tableau, Matplotlib, Seaborn
Database & Querying: SQL (Joins, CTEs, Window Functions), MySQL, PostgreSQL
Advanced Excel: Pivot Tables, Power Query, VLOOKUP/XLOOKUP, Macros (VBA)
Data Manipulation: Data Wrangling, Feature Engineering, Data Transformation
Statistical Techniques: Hypothesis Testing, A/B Testing, Regression Analysis, Probability
Big Data Technologies: Apache Spark (PySpark - Basics), Hadoop (Basics)
ETL & Data Pipelines: ETL Processes, Data Pipelines, Workflow Automation
Model Evaluation: Cross-Validation, Model Performance Metrics (RMSE, Accuracy, Precision, Recall)
Cloud & Tools: AWS (S3, Redshift - Basics), Google BigQuery, Git, GitHub, Jupyter Notebook, VS Code

WORK EXPERIENCE

US Bank, OR | Data Analyst

July 2025 – present

- Analyzed high-volume transaction datasets using **SQL** to identify suspicious activity patterns and anomalies for fraud detection use cases
- Performed **Exploratory Data Analysis (EDA)** using **Python** to uncover fraud trends and customer behavioral patterns across transaction datasets
- Cleaned and preprocessed large-scale financial data, improving data quality and consistency, resulting in 30% better model input reliability
- Visualized fraud insights using **Power BI dashboards**, enabling real-time monitoring and reducing investigation turnaround time by **35%**
- Engineered features and optimized fraud detection models using **scikit-learn**, reducing false positive alerts by 15%
- Implemented scalable data processing and model deployment workflows using **AWS services (Amazon S3, AWS Glue, and AWS Lambda)**, enabling efficient handling of large transaction datasets and improving fraud detection pipeline performance in a cloud environment

Snohomish County | Student Data Scientist (Capstone project)

Jan 2025 – June 2025

- Led Power BI dashboard development for Snohomish County's Low-Income and Disadvantaged Communities (LIDAC) equity analysis system; connected directly to Azure PostgreSQL with custom DAX formulas, dynamic ZIP code map filtering, and real-time response density visualizations — replacing fragmented manual PDF reporting workflows across multiple county departments.
- Designed a four-component interactive dashboard (survey selection, geographic filtering, question-level response analysis, theme visualization) that automated both quantitative and qualitative community survey reporting, eliminated reliance on external consultants, and enabled county decision-makers to access timely equity insights for resource allocation and policy development.

Accenture, India | Data Analyst

Jan 2022 – Aug 2023

- Performed customer segmentation analysis using **Python (pandas)** on retail transaction data, identifying high-value and low-engagement customer groups for targeted strategies.
- Designed targeted customer segmentation strategies for marketing campaigns using **RFM analysis**, improving campaign relevance and audience targeting.
- Developed interactive dashboards using **Microsoft Excel** (Pivot Tables, Power Query, and charts), enabling stakeholders to monitor customer engagement and improving decision-making efficiency by **25%**.
- Built Customer Lifetime Value (CLV) models in **Excel**, prioritizing high-value segments and increasing marketing ROI by **15%**.

EDUCATION

Master's in data science- Seattle University

Integrated MSc Physics and Mathematics- Amrita Vishwa Vidyapeetham

ACADEMIC PROJECTS

Customer Churn Prediction | Python, Scikit-learn, Logistic Regression, Pandas

- Analyzed a telecommunications dataset of 7,043 customers (26.6% churn rate, 21 features) to identify at-risk subscribers before cancellation; performed exploratory data analysis revealing bimodal tenure distribution and differential churn propensity across contract type, service category, and financial burden variables.
- Implemented logistic regression with balanced class weighting to address the 73/27 class imbalance; achieved 80.65% accuracy and 85.2% ROC-AUC; identified contract duration, service type, and customer tenure as dominant churn drivers — producing risk-stratified cohorts that enable targeted retention campaigns before revenue loss occurs.

E-Commerce Data Pipeline with ML Forecasting | Python, Snowflake, dbt Core, Prefect, Prophet, Isolation Forest, Streamlit, Groq API

- Architected a fully automated end-to-end e-commerce data pipeline — synthetic order ingestion into Snowflake via idempotent MERGE statements, dbt transformation through Staging and Mart layers (6 tables, 36 data quality tests all passing), and Prefect daily orchestration chaining ingestion, dbt build, and anomaly detection in sequence across all five pipeline phases.
- Deployed two Prophet forecasting models (30-day revenue and 7-day per-product demand across 20 products), Isolation Forest anomaly detection with severity-coded logging, and a four-page Streamlit dashboard with a Groq-powered AI assistant that generates and executes live SQL against Snowflake — delivering a production-grade system with forecast accuracy tracked over time.